

3D Motion Estimation Using Constant Jerk Model and Kalman Filtering

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Abstract

This report presents the theoretical foundation for estimating the motion of an object in 3D space using Kalman filtering techniques. The system is modeled using a constant jerk motion assumption, this allows accurate tracking of smooth but rapidly changing motion. This report derives the continuous and discrete time state space equations, constructs the state transition and measurement models, and explains both Linear and Extended Kalman Filter formulations.

I. INTRODUCTION

In many real world systems, we can not directly observe the complete state of a moving object. Sensors provide noisy and incomplete measurements, which makes accurate tracking very difficult. Examples of such systems include radar tracking and navigation systems.

To address this problem, state estimation techniques are used to work out the true system state from uncertain measurements. One of the most widely used methods for this purpose is the Kalman Filter. It provides an optimal recursive solution for estimating the state of a linear system under Gaussian noise assumptions.

In this project, the motion of an object in 3D space is modeled using a constant jerk assumption. The state includes position, velocity, acceleration, and jerk along each axis. The objective of this milestone is to develop a complete mathematical model of the system and derive the required equations for applying Linear and Extended Kalman Filters.

This report focuses purely on theoretical modeling and derivation.

II. STATE DEFINITION AND PHYSICAL INTERPRETATION

A. What is a State Vector?

In estimation theory, the *state* of a system is defined as the minimum set of variables that describes the system at a given time. If we know the state of our system at a given time t_0 , and the system dynamics are also known, then the next state of our system can be predicted for any later time t_1 .

In problems related to motion tracking, the state must contain all physical quantities that are necessary to describe how an object moves. This includes not only the current position of the object, but also how that position is changing with time. The Kalman Filter uses the state vector as the central quantity for prediction and correction.

B. Our State Vector

For an object moving in 3D space, we define the state vector as:

$$x = [p_x \ v_x \ a_x \ j_x \ p_y \ v_y \ a_y \ j_y \ p_z \ v_z \ a_z \ j_z]^T \quad (1)$$

This is a 12×1 column vector that includes position, velocity, acceleration, and jerk along each axis.

C. Physical Meaning of Each State Component

1) Position:

$$p(t) = [p_x(t), p_y(t), p_z(t)]^T \quad (2)$$

Meaning: Position represents the location of the object in 3D space at time t .

Units: meters (m).

Purpose: Position is the primary quantity of interest in most tracking systems. Our main objective is to estimate where the object is at any given time.

2) Velocity:

$$v(t) = \frac{dp(t)}{dt} \quad (3)$$

Meaning: Velocity is the rate of change of position with respect to time. It shows how fast the object is moving and in which direction.

Units: meters per second (m/s).

Purpose: Position alone is not sufficient for prediction. Without velocity, it is impossible to estimate where the object will be at the next state.

3) Acceleration:

$$a(t) = \frac{dv(t)}{dt} = \frac{d^2p(t)}{dt^2} \quad (4)$$

Meaning: Acceleration is the rate of change of velocity. It shows the effect of forces acting on the object.

Units: meters per second squared (m/s²).

Purpose: Acceleration allows the model to capture changing velocity, such as speeding up or slowing down. Without acceleration, the system would assume constant velocity, which is unrealistic in many real world scenarios.

4) Jerk:

$$j(t) = \frac{da(t)}{dt} = \frac{d^3p(t)}{dt^3} \quad (5)$$

Meaning: Jerk is the rate of change of acceleration. It describes how smoothly acceleration is changing.

Units: meters per second cubed (m/s³).

Purpose: Jerk is the key attribute in this project. It allows the system to track objects whose acceleration changes over time.

D. Justification for Including Jerk

A common motion model used in tracking systems is the constant acceleration model. The constant acceleration motion assumes that acceleration is fixed over time. This works well for simple motion such as an object moving under constant force or a vehicle moving at constant speed.

However, in real world systems, we do not have constant acceleration. Examples of real world systems include:

- A maneuvering aircraft changing its turn rate
- A vehicle braking or accelerating

In these cases, acceleration itself changes smoothly. The constant jerk model assumes that this change in acceleration is almost constant over small time intervals. This makes the model more realistic for complex motion.

Mathematically, including jerk makes the model a third order kinematic model, since jerk is the third derivative of position. Although this increases the state dimension from 9 to 12, the improvement in tracking accuracy for maneuvering targets makes up for the additional complexity.

III. CONTINUOUS TIME MOTION MODEL

In order to apply Kalman filtering, the physical behavior of the system needs to be described using mathematical equations. These equations define how the state of the system evolves with time. In this project, the motion of the object is modeled using a constant jerk assumption.

A. Differential Equations

Under the constant jerk model, the continuous time dynamics of the system is given by the following differential equations:

$$\frac{dp(t)}{dt} = v(t) \quad (6)$$

$$\frac{dv(t)}{dt} = a(t) \quad (7)$$

$$\frac{da(t)}{dt} = j(t) \quad (8)$$

$$\frac{dj(t)}{dt} = 0 \quad (9)$$

These equations describe the relationship between position, velocity, acceleration, and jerk.

B. Interpretation of the Equations

Each equation has a clear meaning:

Equation (1): Position changes at the rate given by velocity. This is the basic definition of velocity in physics.

Equation (2): Velocity changes at the rate given by acceleration. This captures how the object's speed and direction evolve over time.

Equation (3): Acceleration changes at the rate given by jerk. This allows the model to represent smooth changes in acceleration.

Equation (4): The derivative of jerk is zero, this means that jerk is assumed to be constant with respect to time. This is the main modeling assumption of the system.

Together, these equations form a chain of derivatives, and each state variable is the derivative of the previous one.

C. Understanding the Constant Jerk Assumption

The constant jerk assumption means that over a small time interval Δt , the jerk does not change significantly. This does not mean that jerk is constant for all time. Instead, it means that jerk is treated as a *piecewise constant*. At every time step, the Kalman Filter re-estimates the value of jerk based on new measurements.

In other words, jerk changes slowly over time, but it remains approximately constant between two consecutive states.

D. When is the Assumption Reasonable?

This assumption is reasonable in the following situations:

- When tracking maneuvering targets such as aircraft or vehicles where control inputs change smoothly.
- When the sampling frequency is high enough that jerk appears nearly constant between states.
- When acceleration changes smoothly rather than abruptly.

In such cases, the model provides a good balance between accuracy and mathematical simplicity.

E. When is the Assumption Not Reasonable?

The assumption becomes inaccurate in the following situations:

- When objects experience sudden impacts or collisions, causing instantaneous changes in acceleration.
- When the sampling rate is too low compared to the speed of maneuvering.
- When the object actually follows constant acceleration, in which case jerk becomes unnecessary.

IV. DISCRETE TIME STATE SPACE MODEL

The motion model derived in the previous section is defined in continuous time. However, real world systems operate using discrete measurements. Therefore, the continuous time equations need to be converted into a discrete time form.

A. From Continuous to Discrete Time

A discrete time model is required because:

- Sensor measurements arrive at discrete time instants.
- Computers process data in discrete steps.
- The Kalman Filter algorithm is formulated in discrete time.

Let the sampling interval be denoted by Δt . Our objective is to express the state of the system at time $t + \Delta t$ in terms of the state at time t .

Since the motion along the x, y, and z axes is independent, the derivation can be performed for one dimension and then extended to the rest of the dimensions.

B. Derivation for One Dimension

Starting from the continuous time equations and assuming constant jerk over the interval $[t, t + \Delta t]$, the discrete time equations are derived through integration.

1) *Step 1: Jerk:* Since we are assuming that jerk is constant,

$$j(t + \tau) = j(t), \quad 0 \leq \tau \leq \Delta t \quad (10)$$

2) *Step 2: Acceleration:* Acceleration is the integral of jerk:

$$a(t + \tau) = a(t) + \int_0^\tau j(t) d\tau = a(t) + j(t)\tau \quad (11)$$

3) *Step 3: Velocity:* Velocity is the integral of acceleration:

$$v(t + \tau) = v(t) + \int_0^\tau [a(t) + j(t)\tau] d\tau \quad (12)$$

$$v(t + \tau) = v(t) + a(t)\tau + \frac{1}{2}j(t)\tau^2 \quad (13)$$

4) *Step 4: Position:* Position is the integral of velocity:

$$p(t + \tau) = p(t) + \int_0^\tau \left[v(t) + a(t)\tau + \frac{1}{2}j(t)\tau^2 \right] d\tau \quad (14)$$

$$p(t + \tau) = p(t) + v(t)\tau + \frac{1}{2}a(t)\tau^2 + \frac{1}{6}j(t)\tau^3 \quad (15)$$

5) *Step 5: Evaluate at $\tau = \Delta t$:* Evaluating the above equations at $\tau = \Delta t$ gives:

$$p(t + \Delta t) = p(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 + \frac{1}{6}j(t)\Delta t^3 \quad (16)$$

$$v(t + \Delta t) = v(t) + a(t)\Delta t + \frac{1}{2}j(t)\Delta t^2 \quad (17)$$

$$a(t + \Delta t) = a(t) + j(t)\Delta t \quad (18)$$

$$j(t + \Delta t) = j(t) \quad (19)$$

These equations describe how the state evolves from one time step to the next.

C. Extension to Three Dimensions

Since motion in the x, y, and z axes does not influence each other, the same equations are applied independently to each axis.

For example, along the x-axis:

$$p_x(k + 1) = p_x(k) + v_x(k)\Delta t + \frac{1}{2}a_x(k)\Delta t^2 + \frac{1}{6}j_x(k)\Delta t^3 \quad (20)$$

$$v_x(k + 1) = v_x(k) + a_x(k)\Delta t + \frac{1}{2}j_x(k)\Delta t^2 \quad (21)$$

$$a_x(k + 1) = a_x(k) + j_x(k)\Delta t \quad (22)$$

$$j_x(k + 1) = j_x(k) \quad (23)$$

Identical equations apply for the y and z axes.

D. State Transition Matrix F

The discrete time state space equation can be written in matrix form as:

$$x(k + 1) = Fx(k) \quad (24)$$

For one dimension, the state transition matrix is:

$$F_1 = \begin{bmatrix} 1 & \Delta t & \frac{\Delta t^2}{2} & \frac{\Delta t^3}{6} \\ 0 & 1 & \Delta t & \frac{\Delta t^2}{2} \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (25)$$

For three dimensions, the full matrix is a block diagonal:

$$F = \begin{bmatrix} F_1 & 0 & 0 \\ 0 & F_1 & 0 \\ 0 & 0 & F_1 \end{bmatrix} \quad (26)$$

E. Structure and Interpretation of F

The structure of the matrix F provides important physical insight:

- The matrix is upper triangular, which means higher order states influence lower order states.
- The diagonal entries are equal to one, which shows that each state persists over time.
- The Δt terms show how velocity affects position linearly.
- The $\Delta t^2/2$ terms show how acceleration affects position quadratically.
- The $\Delta t^3/6$ terms show how jerk affects position cubically.

The block diagonal structure indicates the fact that motion along each axis evolves independently. This simplifies implementation.

V. PROCESS NOISE COVARIANCE MATRIX Q

A. What is Process Noise?

In our motion model, we assumed that jerk remains constant between time intervals. This assumption simplifies the mathematics, however, it is not realistic in real world systems. In practice, jerk changes unpredictably due to many unknown forces, pilot maneuvers, turbulence and modeling errors.

These unpredictable effects are referred to as *process noise*. Process noise represents uncertainty in the system dynamics, not in the measurements. Process noise models the fact that the object may not follow the assumed motion equations exactly.

To include this uncertainty, we modify our continuous time model by allowing jerk to change randomly:

$$\frac{dj(t)}{dt} = w(t) \quad (27)$$

where $w(t)$ is white noise.

B. What is White Noise?

White noise represents completely random fluctuations that:

- occur at every instant of time,
- have no memory of past values,
- have zero mean,
- are characterized only by their intensity.

The strength of this noise is described by its *power spectral density* Φ :

$$w(t) \sim \mathcal{N}(0, \Phi) \quad (28)$$

Large values of Φ indicate aggressive random jerk changes, while small values indicate gentle fluctuations.

C. Continuous Time System with Noise

For one axis, the full system becomes:

$$\dot{p}(t) = v(t) \quad (29)$$

$$\dot{v}(t) = a(t) \quad (30)$$

$$\dot{a}(t) = j(t) \quad (31)$$

$$\dot{j}(t) = w(t) \quad (32)$$

In matrix form:

$$\frac{d}{dt} \begin{bmatrix} p \\ v \\ a \\ j \end{bmatrix} = A \begin{bmatrix} p \\ v \\ a \\ j \end{bmatrix} + Gw(t) \quad (33)$$

where:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad (34)$$

The matrix G shows that noise directly affects only jerk and then propagates to other states through integration.

D. How Does Uncertainty Propagate?

If jerk experiences random fluctuations over a time interval Δt , how uncertain do we become about position, velocity, acceleration, and jerk?

This uncertainty growth is captured by the process noise covariance matrix:

$$Q = \int_0^{\Delta t} F(\tau) G \Phi G^T F(\tau)^T d\tau \quad (35)$$

Each component has a clear meaning:

- $F(\tau)$: State transition matrix for time τ ,
- G : Noise input matrix,
- Φ : Noise intensity,
- The integral: Accumulation of noise over time.

E. State Transition Matrix $F(\tau)$

From the motion equations, the continuous transition matrix is:

$$F(\tau) = \begin{bmatrix} 1 & \tau & \tau^2/2 & \tau^3/6 \\ 0 & 1 & \tau & \tau^2/2 \\ 0 & 0 & 1 & \tau \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (36)$$

This reflects how constant jerk propagates through integration.

F. Noise Mapping Term

First compute:

$$G \Phi G^T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \Phi \end{bmatrix} \quad (37)$$

This shows that only jerk receives direct noise.

G. Instantaneous Growth Matrix

Multiplying:

$$M(\tau) = F(\tau) G \Phi G^T F(\tau)^T \quad (38)$$

yields:

$$M(\tau) = \Phi \begin{bmatrix} \tau^6/36 & \tau^5/12 & \tau^4/6 & \tau^3/6 \\ \tau^5/12 & \tau^4/4 & \tau^3/2 & \tau^2/2 \\ \tau^4/6 & \tau^3/2 & \tau^2 & \tau \\ \tau^3/6 & \tau^2/2 & \tau & 1 \end{bmatrix} \quad (39)$$

H. Integration to Obtain Q_1

We now integrate each element from 0 to Δt .

Example:

$$Q_{11} = \int_0^{\Delta t} \Phi \frac{\tau^6}{36} d\tau = \Phi \frac{\Delta t^7}{252} \quad (40)$$

Done for all entries:

$$Q_1 = \Phi \begin{bmatrix} \Delta t^7/252 & \Delta t^6/72 & \Delta t^5/30 & \Delta t^4/24 \\ \Delta t^6/72 & \Delta t^5/20 & \Delta t^4/8 & \Delta t^3/6 \\ \Delta t^5/30 & \Delta t^4/8 & \Delta t^3/3 & \Delta t^2/2 \\ \Delta t^4/24 & \Delta t^3/6 & \Delta t^2/2 & \Delta t \end{bmatrix} \quad (41)$$

I. Full 3D Process Noise Matrix

Since motion in x , y , and z axes is independent:

$$Q = \begin{bmatrix} Q_1 & 0 & 0 \\ 0 & Q_1 & 0 \\ 0 & 0 & Q_1 \end{bmatrix} \quad (42)$$

J. Physical Interpretation

- Position uncertainty grows as Δt^7 (fastest),
- Velocity uncertainty grows as Δt^5 ,
- Acceleration uncertainty grows as Δt^3 ,
- Jerk uncertainty grows linearly.

This occurs because noise enters at the jerk level and integrates upward.

K. Tuning the Parameter Φ

Φ is a design parameter:

- Small Φ : Smooth motion, strong trust in model,
- Large Φ : Aggressive maneuvers, strong trust in measurements.

If Φ is too small, the filter becomes slow and unresponsive. If Φ is too large, the filter becomes noisy and unstable.

L. Why Q is Essential

Without Q , the Kalman Filter assumes perfect dynamics. This leads to overconfidence and poor performance in the real world. The Q matrix allows the filter to remain realistic.

VI. MEASUREMENT MODEL: LINEAR KALMAN FILTER

A. What is a Measurement Model?

The measurement model describes how the internal system state is observed by sensors. The state vector contains many physical quantities, however, in real systems, we cannot measure them directly.

Our state vector is:

$$x = [p_x \ v_x \ a_x \ j_x \ p_y \ v_y \ a_y \ j_y \ p_z \ v_z \ a_z \ j_z]^T \quad (43)$$

B. Measurement Equation

The measurement model is written as:

$$z_k = Hx_k \quad (44)$$

where:

- $z_k \in \mathbb{R}^3$ is the measurement vector,
- $x_k \in \mathbb{R}^{12}$ is the system state,
- $H \in \mathbb{R}^{3 \times 12}$ is the measurement matrix.

C. What Do We Measure?

We assume that the sensor measures only position:

$$z_k = \begin{bmatrix} z_x \\ z_y \\ z_z \end{bmatrix} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad (45)$$

This means:

$$z_x = p_x \quad (46)$$

$$z_y = p_y \quad (47)$$

$$z_z = p_z \quad (48)$$

Velocity, acceleration, and jerk are not directly observable. They are derived by position.

D. Derivation of Measurement Matrix H

Each measurement must be written as a linear combination of all 12 state variables.

1) Row 1: Measurement of p_x :

$$z_x = 1 \cdot p_x + 0 \cdot v_x + 0 \cdot a_x + 0 \cdot j_x + 0 \cdot p_y + 0 \cdot v_y + 0 \cdot a_y + 0 \cdot j_y + 0 \cdot p_z + 0 \cdot v_z + 0 \cdot a_z + 0 \cdot j_z \quad (49)$$

2) Row 2: Measurement of p_y :

$$z_y = 0 \cdot p_x + 0 \cdot v_x + 0 \cdot a_x + 0 \cdot j_x + 1 \cdot p_y + 0 \cdot v_y + 0 \cdot a_y + 0 \cdot j_y + 0 \cdot p_z + 0 \cdot v_z + 0 \cdot a_z + 0 \cdot j_z \quad (50)$$

3) Row 3: Measurement of p_z :

$$z_z = 0 \cdot p_x + 0 \cdot v_x + 0 \cdot a_x + 0 \cdot j_x + 0 \cdot p_y + 0 \cdot v_y + 0 \cdot a_y + 0 \cdot j_y + 1 \cdot p_z + 0 \cdot v_z + 0 \cdot a_z + 0 \cdot j_z \quad (51)$$

E. Final Measurement Matrix

Combining all rows, the measurement matrix is:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \quad (52)$$

F. Physical Interpretation

The structure of H shows that:

- Only position states are observed.
- The filter must derive velocity, acceleration, and jerk indirectly.
- Information flows from measurements through system dynamics.

This model reflects realistic sensors and produces a meaningful estimate.

VII. LINEAR KALMAN FILTER EQUATIONS

A. Introduction and Conceptual Overview

The Linear Kalman Filter is a recursive optimal state estimation algorithm used to estimate the internal state of a dynamic system when the full state cannot be directly measured and measurements are corrupted by noise.

The Kalman Filter combines:

- A prediction obtained from the motion model
 - A measurement obtained from sensors
- to compute the best possible estimate of the state.

The algorithm operates recursively in two repeated steps:

- 1) Prediction (Time Update)
- 2) Update (Measurement Correction)

This recursive structure makes the filter efficient and suitable for real time implementation.

B. Notation and Meaning of Symbols

The Kalman Filter uses the notation:

$$\hat{x}_{n|n-1}$$

which means the estimate of the state at time n , based on all information available up to time $n - 1$. Similarly,

$$\hat{x}_{n|n}$$

represents the estimate at time n after incorporating the measurement at time n .

The hat symbol ($\hat{x}$) denotes an estimate of the true state x .

The matrix P represents the estimation error covariance:

$$P = E[(x - \hat{x})(x - \hat{x})^T]$$

This matrix measures uncertainty in the estimate. Its diagonal elements represent the variance (uncertainty) of each state variable, while the off diagonal elements represent correlations between estimation errors of different state variables.

C. Prediction (Time Update)

The time update step uses the system dynamics to project the previous estimate forward in time.

1) *State Prediction:*

$$\hat{x}_{n|n-1} = F\hat{x}_{n-1|n-1} \quad (53)$$

Where:

- $F \in \mathbb{R}^{12 \times 12}$ is the state transition matrix. It encodes the discrete time constant jerk motion model derived earlier.

This equation predicts where the object should be at time n based on its previous state and the motion equations.

Physically:

- Position is updated using velocity, acceleration, and jerk.
- Velocity is updated using acceleration and jerk.
- Acceleration is updated using jerk.
- Jerk remains constant under the constant jerk assumption.

Thus, the structure of F directly reflects the physical motion model of the object.

2) Covariance Prediction:

$$P_{n|n-1} = F P_{n-1|n-1} F^T + Q \quad (54)$$

Where:

- $P_{n-1|n-1}$ is the previous estimation uncertainty.
- $F P F^T$ propagates uncertainty through system dynamics.
- Q is the process noise covariance matrix.

The term $F P F^T$ describes how previous uncertainty evolves over time due to motion of the system.

In general, uncertainty increases during the prediction step because no new measurements are incorporated.

D. Update (Measurement Correction)

When a new measurement z_k becomes available, the prediction is corrected.

1) Kalman Gain:

$$K_n = P_{n|n-1} H^T (H P_{n|n-1} H^T + R)^{-1} \quad (55)$$

Where:

- H is the measurement matrix.
- R is the measurement noise covariance matrix.
- K_n is the Kalman Gain matrix.

The Kalman Gain determines how much the predicted state should be adjusted based on the measurement.

The term $H P_{n|n-1} H^T$ represents predicted uncertainty expressed in measurement space.

The term $H P_{n|n-1} H^T + R$ represents total uncertainty in the measurement.

If measurement noise R is small:

- Measurements are considered reliable.
- K_n becomes larger.
- The filter relies more on measurements.

If predicted uncertainty $P_{n|n-1}$ is small:

- The model prediction is trusted.
- K_n becomes smaller.
- The filter relies more on prediction.

Thus, the Kalman Gain balances confidence between prediction and measurement.

2) State Update:

$$\hat{x}_{n|n} = \hat{x}_{n|n-1} + K_n (z_n - H \hat{x}_{n|n-1}) \quad (56)$$

The term:

$$z_n - H \hat{x}_{n|n-1}$$

is called the innovation or measurement residual.

It represents the difference between:

- The actual measurement
- The predicted measurement

If this difference is zero, it means the prediction was perfect.

The Kalman Gain scales this innovation to produce the correction term.

Position states are corrected strongly because they are directly measured.

Velocity, acceleration, and jerk are corrected indirectly due to state coupling within the covariance matrix.

3) Covariance Update (Joseph Form):

$$P_{n|n} = (I - K_n H) P_{n|n-1} (I - K_n H)^T + K_n R K_n^T \quad (57)$$

This equation updates the estimation uncertainty after incorporating the measurement.

The first term reduces prior uncertainty.

The second term accounts for uncertainty introduced by measurement noise.

This form is known as the Joseph form and is preferred because it guarantees:

- The covariance matrix remains symmetric.
- The covariance matrix remains positive definite.
- Numerical stability in practical implementations.

After the measurement correction step, uncertainty decreases because new information has been incorporated.

E. Complete Linear Kalman Filter Algorithm

Prediction:

$$\hat{x}_{n|n-1} = F \hat{x}_{n-1|n-1} \quad (58)$$

$$P_{n|n-1} = F P_{n-1|n-1} F^T + Q \quad (59)$$

Update:

$$K_n = P_{n|n-1} H^T (H P_{n|n-1} H^T + R)^{-1} \quad (60)$$

$$\hat{x}_{n|n} = \hat{x}_{n|n-1} + K_n (z_k - H \hat{x}_{n|n-1}) \quad (61)$$

$$P_{n|n} = (I - K_n H) P_{n|n-1} (I - K_n H)^T + K_n R K_n^T \quad (62)$$

The process then repeats for the next time step using $\hat{x}_{n|n}$ and $P_{n|n}$ as the new starting point.

The Linear Kalman Filter provides an optimal linear estimate that minimizes the estimation error covariance under Gaussian noise assumptions. It explicitly handles uncertainty, automatically balances prediction and measurement, and is efficient for real time systems.

VIII. EXTENDED KALMAN FILTER (EKF)

A. Why Do We Need the Extended Kalman Filter?

The Linear Kalman Filter (LKF) assumes that the measurement model is linear:

$$z_n = H x_n \quad (63)$$

This means that the measurement must be a linear combination of the state variables.

However, in many real world systems, measurements are nonlinear functions of the state. Examples include radar, LiDAR, sonar, and vision sensors.

In this milestone, the measurements are given in spherical coordinates:

$$z_n = \begin{bmatrix} r \\ \theta \\ \phi \end{bmatrix}$$

These quantities are nonlinear functions of the Cartesian position (p_x, p_y, p_z) .

There does not exist a constant matrix H such that:

$$z = Hx$$

If we try to apply the Linear Kalman Filter directly:

- The predicted measurement will be incorrect.
- The innovation will not behave properly.
- The filter may diverge.

Therefore, we need the Extended Kalman Filter (EKF).

B. Core Idea of the Extended Kalman Filter

The EKF solves this problem by linearizing the nonlinear measurement function. The nonlinear measurement model is written as:

$$z_k = h(x_k) \quad (64)$$

Instead of using a constant matrix H , the EKF computes a linear approximation of $h(x)$ around the predicted state estimate.

Using first order Taylor expansion:

$$h(x) \approx h(\hat{x}_{k|k-1}) + H_k(x - \hat{x}_{k|k-1})$$

where:

$$H_k = \left. \frac{\partial h(x)}{\partial x} \right|_{x=\hat{x}_{k|k-1}}$$

is the Jacobian matrix evaluated at the predicted state.

Thus, at each time step, the EKF:

- 1) Predicts the state using the linear motion model.
- 2) Computes the Jacobian of the measurement function.
- 3) Uses this Jacobian as the linearized measurement matrix.
- 4) Applies the Kalman update equations.

The key idea is: *The EKF approximates the nonlinear function by a locally linear model at every time step.*

C. Nonlinear Measurement Model in Spherical Coordinates

The measurement function is:

$$h(x) = \begin{bmatrix} r \\ \theta \\ \phi \end{bmatrix}$$

where:

1. *Range*:

$$r = \sqrt{p_x^2 + p_y^2 + p_z^2} \quad (65)$$

Meaning: Distance from the sensor (origin) to the object.

2. *Azimuth*:

$$\theta = \text{atan2}(p_y/p_x) \quad (66)$$

Meaning: Horizontal angle in the x - y plane.

3. Elevation:

$$\phi = \text{atan2} \left(p_z / \sqrt{p_x^2 + p_y^2} \right) \quad (67)$$

Meaning: Vertical angle above or below the horizontal plane.

These functions depend only on position components. Velocity, acceleration, and jerk do not directly affect measurements.

D. Derivation of the Jacobian Matrix

The Jacobian matrix is defined as:

$$H_n = \left. \frac{\partial h(x)}{\partial x} \right|_{x=\hat{x}_{n|n-1}} \quad (68)$$

It contains all first-order partial derivatives.

Definitions: Let:

$$r = \sqrt{p_x^2 + p_y^2 + p_z^2}$$

$$\rho = \sqrt{p_x^2 + p_y^2}$$

Partial Derivatives of Range:

$$\frac{\partial r}{\partial p_x} = \frac{p_x}{r}$$

$$\frac{\partial r}{\partial p_y} = \frac{p_y}{r}$$

$$\frac{\partial r}{\partial p_z} = \frac{p_z}{r}$$

Partial Derivatives of Azimuth:

$$\frac{\partial \theta}{\partial p_x} = -\frac{p_y}{\rho^2}$$

$$\frac{\partial \theta}{\partial p_y} = \frac{p_x}{\rho^2}$$

$$\frac{\partial \theta}{\partial p_z} = 0$$

Partial Derivatives of Elevation:

$$\frac{\partial \phi}{\partial p_x} = -\frac{p_x p_z}{\rho(\rho^2 + p_z^2)}$$

$$\frac{\partial \phi}{\partial p_y} = -\frac{p_y p_z}{\rho(\rho^2 + p_z^2)}$$

$$\frac{\partial \phi}{\partial p_z} = \frac{\rho}{\rho^2 + p_z^2}$$

All derivatives with respect to velocity, acceleration, and jerk are zero.

Complete Jacobian Matrix:

$$H_n = \begin{bmatrix} \frac{p_x}{r} & 0 & 0 & 0 & \frac{p_y}{r} & 0 & 0 & 0 & \frac{p_z}{r} & 0 & 0 & 0 \\ -\frac{p_y}{\rho^2} & 0 & 0 & 0 & \frac{p_x}{\rho^2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{p_x p_z}{\rho(\rho^2 + p_z^2)} & 0 & 0 & 0 & -\frac{p_y p_z}{\rho(\rho^2 + p_z^2)} & 0 & 0 & 0 & \frac{\rho}{\rho^2 + p_z^2} & 0 & 0 & 0 \end{bmatrix}$$

This is a 3×12 matrix evaluated at $\hat{x}_{n|n-1}$.

E. Why the Jacobian is Required

The Jacobian is required because:

- It provides a linear approximation of the nonlinear function.
- It allows uncertainty propagation into measurement space.
- It enables computation of the Kalman gain.

Without the Jacobian, the Kalman update equations cannot be applied to nonlinear systems.

F. Extended Kalman Filter Algorithm

Prediction Step (Same as Linear KF):

$$\hat{x}_{k|k-1} = F\hat{x}_{k-1|k-1} \quad (69)$$

$$P_{k|k-1} = FP_{k-1|k-1}F^T + Q \quad (70)$$

Linearization:

$$H_k = \left. \frac{\partial h(x)}{\partial x} \right|_{x=\hat{x}_{k|k-1}}$$

Predicted Measurement:

$$\hat{z}_k = h(\hat{x}_{k|k-1})$$

Innovation:

$$y_k = z_k - \hat{z}_k$$

Innovation Covariance:

$$S_k = H_k P_{k|k-1} H_k^T + R$$

Kalman Gain:

$$K_k = P_{k|k-1} H_k^T S_k^{-1}$$

State Update:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k y_k$$

Covariance Update:

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^T + K_k R K_k^T$$

IX. SUMMARY AND CONCLUSION

A. Summary of the Work

In this milestone, we developed a complete state estimation framework for tracking a maneuvering object in 3D space using Kalman filtering techniques.

We began by defining a twelve-dimensional state vector that includes:

- Position (p_x, p_y, p_z)
- Velocity (v_x, v_y, v_z)
- Acceleration (a_x, a_y, a_z)
- Jerk (j_x, j_y, j_z)

The inclusion of jerk allows the model to accurately track highly maneuvering targets whose acceleration changes over time.

We derived the discrete time state transition matrix F based on the constant jerk motion model and explained each term in the matrix. We then developed the process noise covariance matrix Q , describing how uncertainty enters the system through unmodeled dynamics.

Next, we formulated the linear measurement model and derived the measurement matrix H as a 3×12 matrix mapping the Cartesian position components of the state to the measurement space.

Since real world sensors often provide measurements in nonlinear forms, we extended the formulation to the Extended Kalman Filter (EKF). We introduced the nonlinear measurement model in spherical coordinates and derived the complete Jacobian matrix required for linearization. This way, the full 3×12 Jacobian matrix was derived.

Finally, we developed the complete EKF algorithm including:

- Prediction step
- Linearization step
- Innovation computation
- Kalman gain calculation
- State update
- Covariance update (Joseph form)

B. Conclusion

This milestone demonstrates a complete theoretical foundation for implementing both the Linear Kalman Filter and the Extended Kalman Filter for tracking applications.

The constant jerk model enables improved tracking performance for highly maneuvering targets by modeling higher order motion dynamics. The EKF formulation allows the system to handle nonlinear measurement models such as spherical coordinates, which are common in radar and sensor systems.

The derivation of the Jacobian matrix is important in enabling linearization of the nonlinear measurement function, making recursive estimation possible even when direct linear measurement models do not exist.

Overall, this work establishes a mathematically rigorous and physically meaningful state estimation framework. It provides the theoretical groundwork required for implementation in assembly language and low level computer architecture environments in the next stages of the project.

X. REFERENCES

- 1) Alex Becker, *Kalman Filter from the Ground Up*, 2023.
- 2) X. R. Li and V. P. Jilkov, "A Jerk Model for Tracking Highly Maneuvering Targets

XI. REPOSITORY LINK

https://github.com/hamzi-codes/CAAL_S26_Project_KalmanFilter_Team-Thunder